

MLDM Assignment 2

Detailed results for Jandson Ribeiro

Course	Machine Learning and Data Mining WS 2022/23 (copy)
Test	MLDM Assignment 2

This are your test results

 Recall

 Properties

Matrix multiplication

Status

Seen but not answered

Your score

0 / 4

0%

Response

Drag and drop the properties of matrix multiplication to its current answer.

A, B and C are Matrices

I is the Identity Matrix

Assuming that the dimensions of matrices are compatible and allow matrix multiplication

Note:

Correct selection - (+1) ; Wrong - (-1) ; Overall min. mark of a question is always 0

AB = BA

A (B+C) = AB + AC

IA = AI = A

A (BC) = (AB) C

Associative Property

Identity Property

Distributive

Does not hold in matrix multiplication

<https://olat.vcrp.de/auth/RepositoryEntry/3909322134/CourseNode/106528572880292/test/0>

1/10

Transpose Matrix

☰ Transpose

Status Not started

Your score 0 / 5 0%

Response

Choose right or wrong for every statement.

Given ,

$\mathbf{A}$ is a Matrix ; $\mathbf{A}^T$ is the transpose of Matrix $\mathbf{A}$

$\mathbf{I}$ is the Identity Matrix

Note:

Correct selection - (+1) ; Wrong - (-1) ; Overall min. mark of a question is always 0

Unanswered	Right	Wrong	
☐	☐	☐	If a matrix $\mathbf{A}$ is of order $m \times n$, then the transpose of the matrix $\mathbf{A}^T$ is also of the order $m \times n$.
☐	☐	☐	The transpose of an invertible matrix is also invertible, and its inverse is actually the transpose of the inverse of the original matrix.
☐	☐	☐	Transpose when applied to a matrix, multiplication and addition operations have the higher precedence than Transpose
☐	☐	☐	A square matrix $\mathbf{A}$ of order $n \times n$ is considered to be an orthogonal matrix. It is possible only when $\mathbf{A}^T \times \mathbf{A} = \mathbf{A} \times \mathbf{A}^T = \mathbf{I}$
☐	☐	☐	The determinant of a matrix is not zero if and only if the matrix is invertible and the linear map represented by the matrix is not an isomorphism

Orthogonality

☰ Orthogonal

Status Not started

Your score 0 / 6 0%

Response

Two vectors $\mathbf{x}$, $\mathbf{y}$ said to be orthogonal when:

Note:

Correct selection - (+1) ; Wrong - (-1) ; Overall min. mark of a question is always 0

Unanswered	Right	Wrong	
☐	☐	☐	$\mathbf{y}$ is perpendicular to $\mathbf{x}$
☐	☐	☐	$\mathbf{x}$ is a zero vector
☐	☐	☐	$\mathbf{x}$ is parallel to $\mathbf{y}$
☐	☐	☐	$\mathbf{y}$ is a zero vector
☐	☐	☐	$\mathbf{x} \cdot \mathbf{y}$ is 0

*** Vector orthogonality

Status

Not started

Your score

0 / 2

0%

Response

Are these vectors orthogonal?
Give your answer as 'yes' or 'no' .
a. $(1,3,2)^T, (4,-1,0)^T, (2,-1,0)^T$ -
b. $(1,3,-5)^T, (1,3,2)^T, (3,-1,0)^T$ -

 Vector operations

☰ Dot product

Status

Not started

Your score

0 / 4

0%

Response

Choose right or wrong for the following statments.
 a, b are vectors
 $a.b$: dot product of vector a and b
 c : constant
Note:
Correct selection - (+1) ; Wrong - (-1) ; Overall min. mark of a question is always 0

Unanswered	Right	Wrong	
☐	☐	☐	$a.a = 2a$
☐	☐	☐	$(ca) \cdot b = c(a \cdot b)$
☐	☐	☐	$a.b = a b \cos \theta$
☐	☐	☐	$a.b \neq b.a$

☰ Cross product

Status

Not started

Your score

0 / 4

0%

Response

Choose right or wrong for the following statments about Cross product.
 a, b are vectors
 $a \bullet b$, and $a \times b$: crossproduct of vector a and b
Note:
Correct selection - (+1) ; Wrong - (-1) ; Overall min. mark of a question is always 0

Unanswered	Right	Wrong	
☐	☐	☐	$a \times (b \bullet c) \neq (a \bullet b) \times c$
☐	☐	☐	$a \bullet b \perp a, b$

Knowledge questions

Similarity metrics

Status Not started

Your score 0 / 5 0%

Response

Choose right or wrong for each statement.

Note:

Correct answer - (+1) ; Wrong - (-1) ; Overall min. mark of a question is always 0

Unanswered	Right	Wrong	
☐	☐	☐	Jaccard Similarity is a measure to find similarity between two sets of sample datasets.
☐	☐	☐	Cosine similarity is used to find similarity in the subject between two documents or datasets.
☐	☐	☐	Jaccard similarity is the ratio of size of intersection over union of the datasets
☐	☐	☐	Mahalanobis distance is a measure to find similarity between 2 distributions
☐	☐	☐	While dealing with 2 datasets, the distance between them can be takes as the value of similarity

☒ Distance metric

Status Not started

Your score 0 / 2 0%

Response

Which of the following options are not a distance metric?

- ☐ Hamming
- ☐ Cosine
- ☐ Minkowski
- ☐ Wasserstein

Class imbalance

Status Not started

Your score 0 / 8 0%

Response

What is a class imbalance problem? How can it be solved? Support your answer with an example.

0 word

 Missing values

Status Not started

Your score 0 / 5 0%

Response

Missing values can be filled with the following values

Note:
Correct selection - (+1) ; Wrong - (-1) ; Overall min. mark of a question is always 0

Unanswered	Right	Wrong	
☐	☐	☐	mean of the corresponding attribute of all samples
☐	☐	☐	mode of corresponding attribute when we have extreme values
☐	☐	☐	most frequent value in categorical attribute
☐	☐	☐	random sampling of the corresponding attribute
☐	☐	☐	mean of the corresponding attribute os samples belonging to the same class

☒ Feature scaling

Status Not started

Your score 0 / 3 0%

Response

Why does feature scaling not affect algorithms like decision tree and random forest?

☐ Feature scaling cannot be applied on these algorithms

☐ These algorithms do not use features

☐ These algorithms are not sensitive to variance in data

☐ These algorithms are distance based

☐ Outlier removal

Status Not started

Your score 0 / 2 0%

Response

What is a loss function in machine learning?

☐ identifies how many outliers are present in the data

☐ estimates the probability of a data point being an outlier

☐ calculates how far a true value is from an estimated value

☐ calculates the loss caused by the outlier removal





Evaluation

Manhattan distance

Status

Not started

Your score

0 / 2

0%

Response

Calculate the **Manhattan distance (d)** between the 2 points **x** and **y**

$$x = \begin{pmatrix} 3 \\ 9 \\ 2 \end{pmatrix}$$
$$y = \begin{pmatrix} 6 \\ 5 \\ 13 \end{pmatrix}$$

d =

Euclidean distance

Status

Not started

Your score

0 / 2

0%

Response

Calculate the **Euclidean distance (d)** between the 2 vectors **x** and **y**.

The answer has to be in the decimal number format with up to 2 decimal points.

$$x = \begin{pmatrix} 12 \\ 9 \\ 14 \end{pmatrix}$$
$$y = \begin{pmatrix} 15 \\ 13 \\ 3 \end{pmatrix}$$

d =

 Error Functions

Mean Absolute Error (MAE)

Status

Not started

Your score

0 / 4

0%

Response

You are provided with the following data:

y_original	y_predicted
3.88	5.72
4.55	6.37
5.22	7.01
5.88	7.66
6.56	8.30
7.22	8.95
7.89	9.60
8.56	10.25

Calculate the following:

The Mean Absolute Error:

Note:

- The answer can have up to 2 decimal places

Mean Square Error (MSE)

Status

Not started

Your score

0 / 4

0%

Response

You are provided with the following data:

y_original	y_predicted
3.88	5.72
4.55	6.37
5.22	7.01
5.88	7.66
6.56	8.30
7.22	8.95
7.89	9.60
8.56	10.25

Calculate the following:

The Mean Squared Error:

Note:

- The answer can have up to 2 decimal places

Root Mean Squared Error (RMSE)

Status

Not started

Your score

0 / 4

0%

Response

You are provided with the following data:

y_original	y_predicted
3.88	5.72
4.55	6.37
5.22	7.01
5.88	7.66
6.56	8.30
7.22	8.95
7.89	9.60
8.56	10.25

Calculate the following:

The Root Mean Squared Error:

Note:

- The answer can have up to 2 decimal places

 Missing values

...

Missing value

Status

Not started

Your score

0 / 4

0%

Response

In the following table, fill in the missing values by using statistical methods.

x1, x2 and x3 are features

ω is the class .

For missing value in x2, fill with mean of the corresponding attribute belonging to the same class.

For missing value in x3, fill with the mean of the corresponding attribute of all samples.

x1	x2	x3	ω
green	10	7	0
blue	23	2	1
green	2		0
red	6	9	1
yellow		5	1
	4	7	1
red		4	0
blue	5	6	1
blue	1	2	0
grey	7	7	1
blue	20	8	0

 Programming

 Essay		
Status	Not started	
Your score	0 / 30	0%

Response

First download the file dataset.csv

The record contains row-wise samples of the fruits Apple, Lemon, and Pear. The attributes are listed in the first three columns, the class (Fruit) is provided in the forth column.

Implement below described processing tasks in a way that you may input another dataset with identical structure but different values. Print the processed dataset after each step onto your submission.

Write the following functions.

a. To delete duplicate samples. -

- **def del_duplicate(data: dataframe) -> returns dataframe**

b. Function to fill missing values (fields that have 0) with the mean of the corresponding class of the attribute

- **def fill_values(data: dataframe) -> returns dataframe**

c. Function to transform nominal attributes (not the classes!) into numerical values.

For example, there is an attribute called "color" and it is assigned values of "green", "yellow", and "red". Remove this "color" column and introduce three new numerical columns, "green", "yellow", and "red". For each row, set the value of the column corresponding to the former nominal value to one and the other two to zero.

- **def transform_nominal(data: dataframe) -> returns dataframe**

d. Function to normalise the values of each numerical attribute with min-max method

- **def normalise_minmax(data: dataframe) -> returns dataframe**

Also write the output of each function to a csv file.

Note:

- **If the code has any syntactical, logic errors, and/or exceptions it will not be accepted.**

0 word